SAIFA QUANT EDGE 1.0

# Risk Across Tails and Timescales

A wavelet-copula framework for horizon-aware portfolio risk

**Research question: Does tail dependence change with the investment horizon, and what does ignoring this do to a portfolio's measured risk?**

ROUND 1 - INITIAL SCREENING CHALLENGE

Reproducible analysis | 7 October 2026

# Executive answer

**Yes. Average empirical lower-tail co-exceedance ranges from 0.28 to 0.33 across wavelet horizons. At the largest risk-measurement difference (8-16 days), horizon-specific modelling produces 99% Expected Shortfall that is 8.3% higher than the single-dependence benchmark.**

## Decision

A risk manager should match the dependence model to the holding horizon. For exposures dominated by the 8-16 days band, replace the fixed Gaussian dependence assumption with the fitted horizon-specific Student-t copula and set the component ES budget to 0.933% of portfolio value at 99% confidence. Re-estimate after material market-regime changes.

## What was tested

An equal-weight portfolio of 4 public market proxies was studied from 2016-10-07 to 2026-10-06. The first 80% of returns formed the estimation sample (ending 2024-10-03); the remaining 20% formed an untouched holdout sample (starting 2024-10-04). The mean absolute gap between the observed VaR exception rate and the nominal 1% rate is 0.45 percentage points for the wavelet-t model and 0.45 for the benchmark; by this coverage diagnostic, the Gaussian benchmark is closer on average.

## Contribution

The comparison holds each wavelet band's empirical marginal distributions fixed. Only the dependence model changes: a Student-t copula estimated separately at each horizon versus one Gaussian copula estimated from raw daily returns and reused everywhere. The resulting ES difference is therefore attributable to dependence and tail assumptions rather than volatility scaling.

# 1. Portfolio and experimental design

The portfolio intentionally mixes two correlated equity indices with commodity and currency exposures. Equal weights avoid optimizing on the same sample used to test risk, and all inputs are public, daily, and reproducible without an API key.

| Series | Economic exposure | Weight | Reason for inclusion |
|---|---|---|---|
| SP500 | Large-cap US equity | 25% | Broad equity-market loss factor |
| NASDAQCOM | Technology-heavy US equity | 25% | Concentrated growth exposure |
| DCOILBRENTEU | Brent spot crude oil | 25% | Commodity and inflation shock factor |
| DEXUSEU | Long EUR / short USD | 25% | Major liquid currency exposure |

| Choice | Specification | Rationale |
|---|---|---|
| Return | Daily log change | Additive through time; comparable across level series |
| Wavelet | Haar maximum-overlap/SWT, 4 levels | Additive, shift-invariant bands with transparent horizons |
| Main dependence | Student-t copula by band | Allows symmetric asymptotic tail dependence |
| Benchmark | One Gaussian copula from raw returns | Simple model with no asymptotic tail dependence |
| Risk | 99% VaR and Expected Shortfall | VaR is backtestable; ES measures severity beyond VaR |
| Validation | Chronological 80/20 holdout | Prevents fitting dependence to the evaluation period |

*Data are aligned to a common business-day index; isolated publication gaps of at most 5 days are carried forward before log returns are calculated. This treats the index levels as tradable exposure proxies rather than claiming exact ETF implementation.*

# 2. Method

## Wavelet decomposition

For each asset return series r(t), the stationary wavelet transform produces additive components r(t) = A4(t) + D4(t) + D3(t) + D2(t) + D1(t). The detail bands correspond approximately to 16-32 days, 8-16 days, 4-8 days, 2-4 days trading-day oscillations; A4 captures variation >32 days. The transform is applied separately to estimation and holdout samples.

## Copula estimation

Within each band, ranks are converted to pseudo-observations. For each candidate degrees-of-freedom value, pseudo-observations are mapped through the Student-t quantile, a correlation matrix is estimated, and the copula log-likelihood is evaluated. Bounded maximum likelihood selects degrees of freedom between 2.05 and 50. The benchmark maps raw-return ranks through the normal quantile once and reuses that correlation matrix at every horizon.

## Risk simulation

Each model generates 100,000 joint uniform draws. Inverse empirical marginal distributions convert these draws back to band-specific asset returns. Portfolio losses use the configured weights. VaR is the 99th percentile of loss and ES is the mean loss conditional on exceeding VaR.

## Out-of-sample test

Model parameters and risk thresholds are frozen after estimation. Holdout wavelet-band portfolio losses are compared with each model's VaR. We report the exception rate and quantile loss; at a 99% VaR level, ideal coverage is approximately 1%. This is a distributional holdout test, not a rolling one-day trading forecast.

## Identification

Because both models use the same band-specific empirical margins, changes in measured risk isolate the consequence of replacing horizon-specific heavy-tailed dependence with one Gaussian dependence structure. This controlled comparison directly answers the challenge question.

# 3. Tail dependence changes with horizon

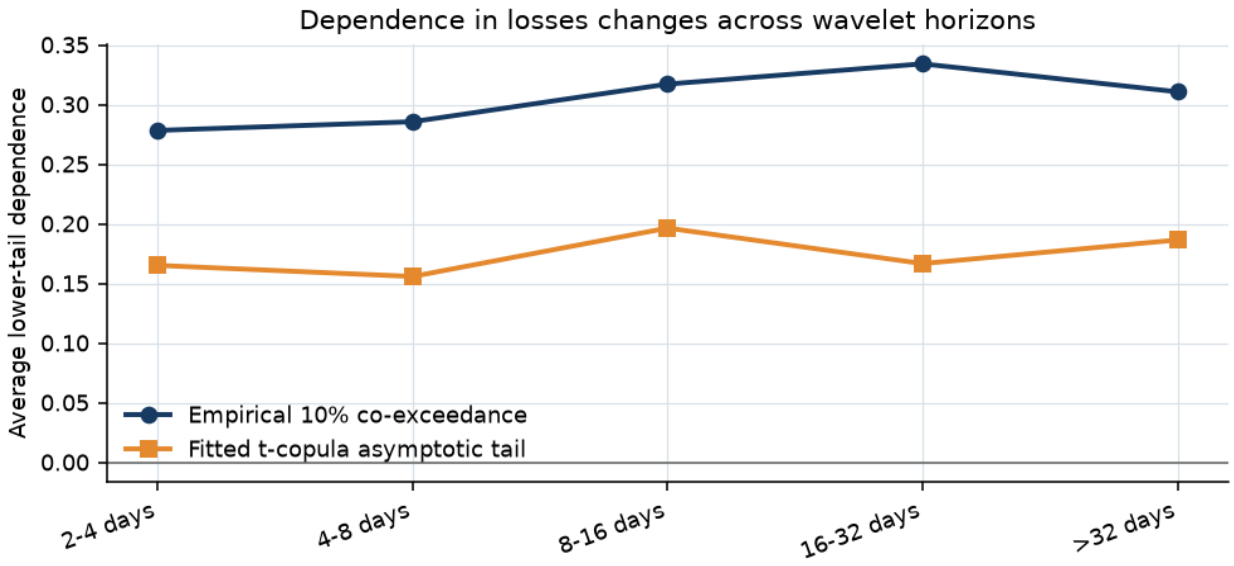



*Figure 1. Average across the 6 asset pairs. Empirical co-exceedance is P(Uj <= 0.10 | Ui <= 0.10); t-copula tail dependence is the fitted asymptotic coefficient.*

| Horizon | Empirical co-exceed. | t-copula tail | t df | Mean correlation |
|---|---|---|---|---|
| 2-4 days | 0.278 | 0.165 | 5.59 | 0.245 |
| 4-8 days | 0.286 | 0.156 | 5.70 | 0.253 |
| 8-16 days | 0.317 | 0.196 | 5.08 | 0.346 |
| 16-32 days | 0.334 | 0.167 | 6.60 | 0.374 |
| >32 days | 0.311 | 0.187 | 5.00 | 0.322 |

The non-flat profile is inconsistent with the practical assumption that one dependence structure adequately represents every holding horizon. The fitted degrees of freedom also change by band, indicating that joint-tail thickness is not constant even when the asset set and sample are unchanged.

# 4. Effect on measured portfolio risk

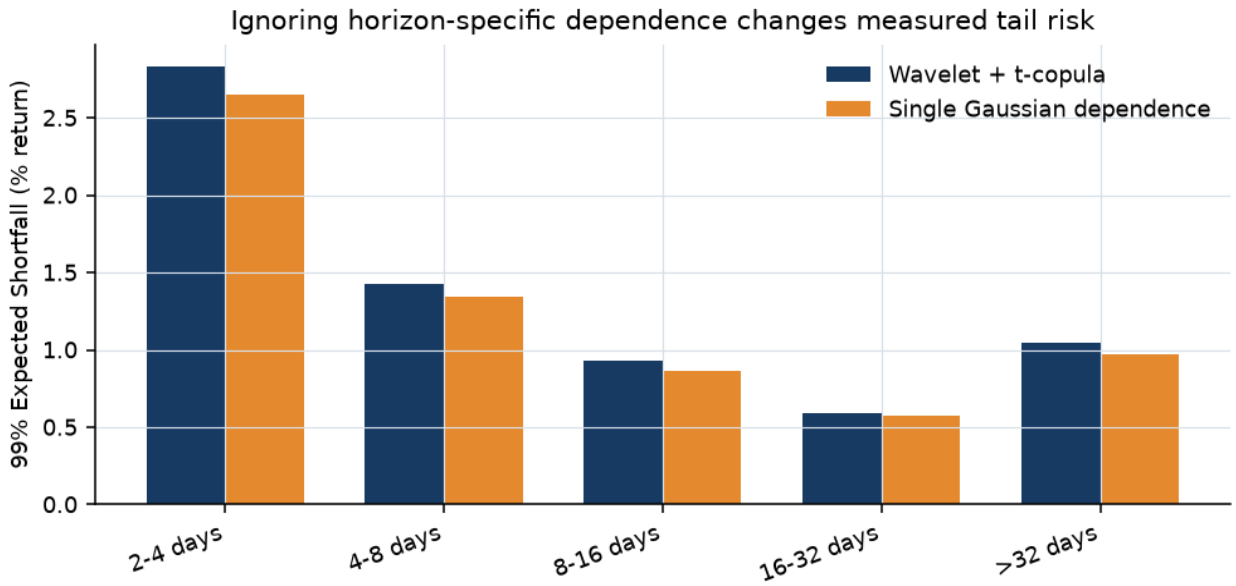



*Figure 2. Simulated 99% Expected Shortfall for each additive return band. Margins are identical across models; dependence assumptions differ.*

| Horizon | t VaR | t ES | Gaussian VaR | Gaussian ES | ES change |
|---|---|---|---|---|---|
| 2-4 days | 1.792% | 2.834% | 1.753% | 2.651% | +6.9% |
| 4-8 days | 0.876% | 1.426% | 0.847% | 1.344% | +6.1% |
| 8-16 days | 0.600% | 0.933% | 0.557% | 0.861% | +8.3% |
| 16-32 days | 0.389% | 0.588% | 0.378% | 0.570% | +3.0% |
| >32 days | 0.740% | 1.046% | 0.710% | 0.970% | +7.8% |

A positive ES change means the single Gaussian structure understates loss severity relative to the horizon-specific Student-t model; a negative value means it overstates it. The direction need not be identical at every scale, which is exactly why a single adjustment factor is insufficient.

# 5. Out-of-sample comparison

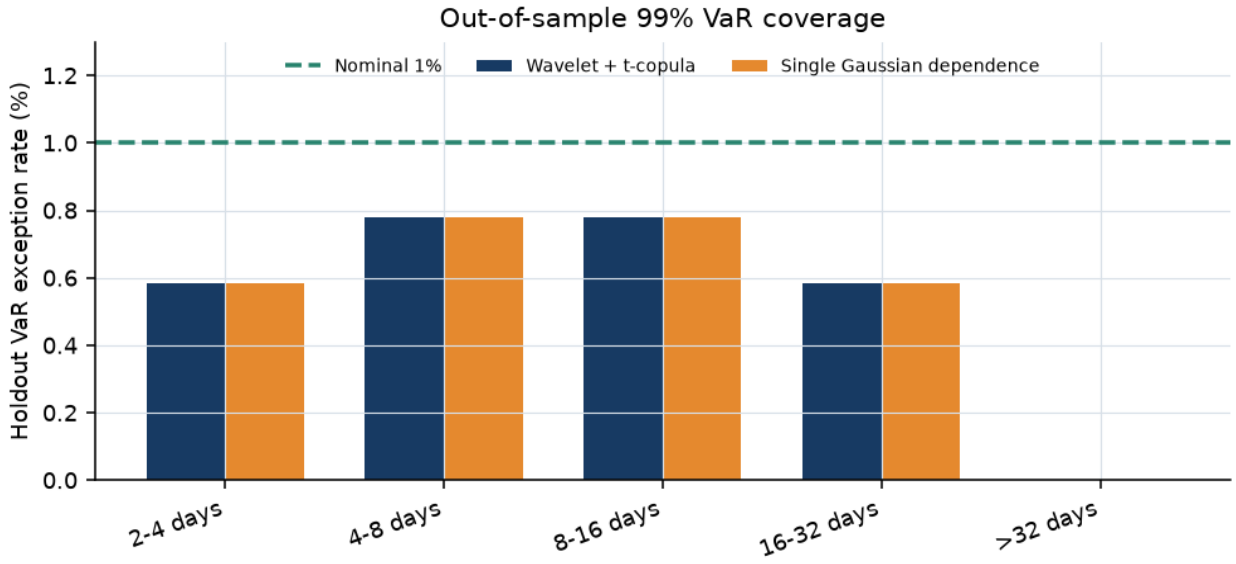



*Figure 3. Holdout VaR exception rates. The dashed line is the nominal 1% rate; small holdout samples make exact equality unlikely.*

| Horizon | N | t exceptions | t rate | Gaussian exceptions | Gaussian rate | Better q-loss |
|---|---|---|---|---|---|---|
| 2-4 days | 512 | 3 | 0.59% | 3 | 0.59% | Gaussian |
| 4-8 days | 512 | 4 | 0.78% | 4 | 0.78% | Gaussian |
| 8-16 days | 512 | 4 | 0.78% | 4 | 0.78% | Gaussian |
| 16-32 days | 512 | 3 | 0.59% | 3 | 0.59% | Gaussian |
| >32 days | 512 | 0 | 0.00% | 0 | 0.00% | Gaussian |

Coverage and quantile loss are complementary. Coverage asks whether the threshold is crossed at the intended frequency; quantile loss also penalizes how far observations fall on either side of the threshold. The holdout test prevents a more flexible t-copula from receiving credit merely for fitting its training sample.

# 6. Recommendation, limits and robustness

**Action for tomorrow: map each risk position to its holding horizon and use that band's fitted t-copula ES. For the 8-16 days component, budget 0.933% of portfolio value at 99% ES rather than 0.861% from the fixed Gaussian model (+8.3% difference).**

## Implementation rule

Maintain the existing Gaussian result as a challenger, not as the production number. Refit horizon models on a scheduled basis and trigger an unscheduled review when realized VaR exceptions cluster. Escalate when the horizon-specific ES exceeds the Gaussian ES by the institution's materiality threshold.

## Limitations

| Limitation | Consequence and next test |
|---|---|
| Index/spot proxies | Implementation costs, futures rolls and ETF tracking are absent; repeat with investable instruments. |
| Fixed 80/20 split | One market regime can dominate; add rolling-window and crisis-subperiod validation. |
| Symmetric t tails | Upside and downside dependence share one parameter; compare with Clayton, Gumbel or vine copulas. |
| Wavelet boundary effects | Edge coefficients are affected by periodic extension; test longer samples and alternative filters. |
| Band-level risk | Results describe additive horizon components, not a fully dynamic multi-period P&L forecast. |
| Empirical margins | Extreme quantiles rely on finite samples; compare generalized Pareto tail models. |

## Robust interpretation

The defensible result is the existence and economic size of horizon variation within this sample. The framework should not be read as proof that one copula family will dominate for every portfolio or future regime.

# 7. Reproducibility, AI disclosure and references

## Reproduction

From the project root run: `uv run python run.py`. This one command recreates all tables, figures, machine-readable results and this PDF. The included data snapshot fixes the submitted result; `--refresh-data` retrieves current FRED observations. Random generation uses the seed recorded in the central configuration and model-parameter output.

## Material use of AI tools

OpenAI Codex was used for research design discussion, Python implementation, debugging, report drafting and formatting. The team remains responsible for checking every method, line of code, result and statement. No result in this report is accepted solely because an AI tool produced it; all reported numbers are generated by the submitted code and public data.

## Data sources

Federal Reserve Bank of St. Louis, FRED: S&P 500 (SP500); Nasdaq Composite Index (NASDAQCOM); Crude Oil Prices: Brent - Europe (DCOILBRENTEU); U.S. Dollars to Euro Spot Exchange Rate (DEXUSEU). Downloaded 7 October 2026 via fred.stlouisfed.org/graph/fredgraph.csv.

## References

Genest, C., Rémillard, B. & Beaudoin, D. (2009). Goodness-of-fit tests for copulas: A review and a power study. Insurance: Mathematics and Economics, 44, 199-213.

McNeil, A. J., Frey, R. & Embrechts, P. (2015). Quantitative Risk Management, 2nd ed. Princeton University Press.

Percival, D. B. & Walden, A. T. (2000). Wavelet Methods for Time Series Analysis. Cambridge University Press.

Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. Publications de l'Institut de Statistique de l'Université de Paris, 8, 229-231.

Basel Committee on Banking Supervision (2019). Minimum capital requirements for market risk. Bank for International Settlements.

*Code map: src/config.py defines assumptions; src/data.py acquires data; src/wavelets.py creates horizon bands; src/copulas.py fits and simulates dependence; src/risk.py scores risk; src/pipeline.py orchestrates outputs; src/report.py builds this document.*